

(c) The main program:

- attempts to store each of the integers 1 to 25 (inclusive) in the queue in ascending numerical order using `Enqueue()`
- outputs the integer that was passed to `Enqueue()` and "Successful" if it was stored in the queue, or "Unsuccessful" if it was **not** stored in the queue.

For example:

- if the integer 5 is passed to `Enqueue()` and is stored in the queue, the output will be: "5 Successful"
- if the integer 23 is passed to `Enqueue()` and is **not** stored in the queue, the output will be: "23 Unsuccessful"

Write program code for the main program.

Save your program.

Copy and paste the program code into part 1(c) in the evidence document.

[3]

- (d) The function `Dequeue()` returns -1 if the queue is empty. If the queue is **not** empty, the function returns the next item in the queue, updates the appropriate pointers and updates `NumberItems`

Write program code for `Dequeue()`

Save your program.

Copy and paste the program code into part 1(d) in the evidence document.

[6]

- (e) (i) Write program code to extend the main program to call `Dequeue()` twice and output the return value each time.

Save your program.

Copy and paste the program code into part 1(e)(i) in the evidence document.

[2]

- (ii) Test your program.

Take a screenshot of the output(s).

Save your program.

Copy and paste the screenshot into part 1(e)(ii) in the evidence document.

[1]